

Learning Validator Archetypes from On-Chain Solana Data

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Abstract

Blockchain validators play a critical role in maintaining the security, reliability, and performance of proof-of-stake networks. Despite their importance, validators are often treated as a homogeneous population, potentially overlooking meaningful differences in operational behavior, performance, and economic strategy. This study investigates whether distinct validator archetypes can be identified directly from on-chain Solana data using unsupervised learning techniques.

Publicly available validator data from the Solana blockchain was obtained through Google BigQuery's public blockchain datasets. Following extensive exploratory analysis of the 125 GB source dataset, validator vote accounts were identified as the primary unit of analysis. A validator-level dataset was constructed using features related to commission policy, operational scale, performance, consistency, and historical activity.

Principal Component Analysis (PCA) was applied to reduce dimensionality while preserving the majority of variation in the data. Four principal components explaining approximately 85% of total variance were retained and used as inputs to a K-Means clustering model. Cluster selection was guided by both the Elbow Method and Silhouette Analysis, resulting in a six-cluster solution.

The analysis identified six distinct validator archetypes within the Solana ecosystem: Low-Commission Mature Validators, High-Commission Mature Validators, Emerging Validators, Inactive Validators, Volatile Validators, and High-Performance Validators. These groups differed across multiple dimensions, including validator maturity, performance quality, consistency, operational scale, and commission strategy.

The results suggest that the Solana validator population is far from homogeneous and that meaningful structure can be extracted directly from publicly available on-chain data. More broadly, this work demonstrates the value of statistical learning techniques for understanding complex blockchain ecosystems and establishes a foundation for future research into validator behavior, stake dynamics, and network performance.

1. Introduction

Blockchain networks rely on decentralized validators to verify transactions, participate in consensus, and maintain the security of the network. In the Solana ecosystem, validators are responsible for voting on blocks and helping the network reach consensus while supporting high transaction throughput and low latency. Because validators collectively determine the performance, reliability, and decentralization of the network, understanding their behavior is important for both researchers and network participants.

While considerable attention has been given to blockchain performance, staking economics, validator incentives, and consensus mechanisms, comparatively little work has focused on identifying distinct validator archetypes from on-chain data. Validators are often treated as a homogeneous population despite substantial differences in operational scale, commission policies, voting performance, and historical activity. It is reasonable to expect that different categories of validators exist, ranging from large established operators to smaller emerging participants. Identifying these groups may provide insight into the economic and operational structure of the Solana network.

This project was motivated by a broader interest in blockchain analytics and ongoing research involving the Solana ecosystem. As blockchain networks continue to grow in scale and complexity, understanding the characteristics of their participants becomes increasingly important. Rather than beginning with predictive modeling, this study focuses on a more fundamental question: what types of validators exist within the Solana network, and can those validator archetypes be identified directly from on-chain data?

To answer this question, validator vote account data from the Solana blockchain was analyzed and transformed into a validator-level dataset containing features related to operational scale, commission policy, historical activity, and voting performance. Unsupervised learning techniques were then applied to discover patterns and group validators according to their behavioral and operational characteristics.

The analysis identified several distinct validator archetypes within the Solana ecosystem, including Low-Commission Mature Validators, Emerging Validators, High-Commission Mature Validators, High-Performance Validators, Inactive Validators, and Unstable Validators. These findings suggest that the validator population is far from homogeneous and that meaningful structure can be extracted directly from on-chain data. More broadly, this work demonstrates how statistical learning techniques can be used to improve our understanding of complex blockchain ecosystems and provides a foundation for future research into validator behavior and network dynamics.

2. Data

This project uses publicly available on-chain data from the Solana blockchain, accessed through Google's BigQuery Public Datasets program. The primary source of data was the [*crypto_solana_mainnet_us.Accounts*](#) table, which contains account-level information collected directly from the Solana mainnet. Google aggregates and maintains this dataset to support large-scale analytics and research on blockchain networks.

The Solana dataset was selected for two primary reasons. First, Solana is one of the largest proof-of-stake blockchain networks, making it an interesting environment for studying validator behavior and network structure. Second, this project was motivated by my ongoing research involving the Solana ecosystem, creating an opportunity to develop analyses that may contribute to future research efforts beyond the scope of this course project.

The original BigQuery dataset exceeded 125 GB in size and contained hundreds of millions of blockchain account records. Because this course focuses on big data analytics, the full dataset was used extensively during the exploratory phase of the project. Initial investigations involved querying the complete dataset to understand its schema, identify relevant account types, examine account distributions, and evaluate the feasibility of different research directions. These exploratory queries guided feature selection and dataset construction.

Several alternative Solana datasets were also considered during the project planning stage, including publicly available Solana archives hosted through Hugging Face. While these datasets contained large amounts of blockchain information, they lacked validator vote transactions, which are essential for evaluating validator behavior and performance. As a result, the Google BigQuery dataset was ultimately chosen because it provided direct access to validator vote accounts, voting history, commission information, stake-related information, and epoch credit records.

Since the objective of this study was to identify validator archetypes, the analysis was restricted to validator vote accounts. Exploratory analysis revealed approximately 8,500 unique validators represented by roughly 9,300 vote account records. Nearly all validators maintained a single vote account throughout their recorded history, making vote accounts an effective proxy for individual validators. The dataset also contained over 155 million records of associated epoch credit histories, providing valuable information on validator performance and historical activity.

This dataset is particularly valuable because it provides a direct record of validator activity recorded on-chain rather than relying on surveys, self-reported information, or third-party estimates. Of particular importance are Solana's epoch credit records, which reward validators for successfully participating in consensus and accurately voting on the canonical blockchain fork. As a result, epoch credits serve as a useful proxy for validator performance and reliability over time. Combined with information on stake, commission rates, voting history, and operational longevity, these records provide a comprehensive view of validator behavior. This

makes the dataset especially well suited for unsupervised learning, where the objective is to discover previously unknown structure and identify meaningful validator archetypes within a large and complex blockchain ecosystem.

3. Methodology

The objective of this study was to identify distinct validator archetypes within the Solana ecosystem using unsupervised learning techniques. Because the structure of the dataset and the relationships between blockchain entities were not known beforehand, the project began with an exploratory phase focused on schema analysis and scope definition.

Initial analysis was conducted on the full 125 GB Solana Accounts dataset to identify relevant account types and determine how validators were represented on-chain. Several account categories, including vote accounts, stake accounts, and delegated accounts, were examined to better understand their relationships and potential usefulness for modeling. Through this process, validator vote accounts were identified as the most appropriate unit of analysis, and epoch credits were selected as the primary performance-related signal due to their direct connection to validator voting behavior and consensus participation.

After filtering the dataset to validator vote accounts, duplicate validator records were removed and exploratory data analysis was performed to evaluate feature distributions, missing values, and potential outliers. Features that provided little information or were largely redundant were excluded from further analysis.

The final modeling dataset consisted of seven features. One original feature, validator commission rate, was retained directly from the source data. Six additional features were engineered to capture key dimensions of validator behavior: account balance (log-transformed lamports) as a measure of validator scale, median efficiency percentage as a measure of typical performance, efficiency coefficient of variation as a measure of consistency, performance trend as a measure of recent improvement or decline, number of epochs recorded as a measure of operational longevity, and a binary history indicator identifying whether sufficient historical records were available for a validator.

Following feature engineering, missing values were imputed where appropriate and all numerical features were standardized to ensure comparability across different scales. Principal Component Analysis (PCA) was then applied to reduce dimensionality while preserving the majority of variation present in the data. Four principal components were retained, capturing approximately 85% of the total variance.

Finally, K-Means clustering was performed on the principal component representation of the data. The optimal number of clusters was determined using both the elbow method and

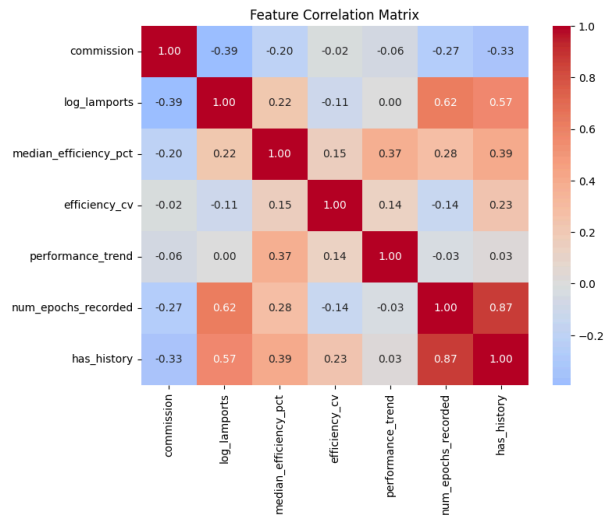
silhouette analysis. Based on these criteria, a six-cluster solution was selected and used to identify and characterize distinct validator archetypes within the Solana network.

4. Results

4.1. Feature Relationships and Data Structure

Before applying dimensionality reduction and clustering, relationships between the engineered validator features were examined using a correlation matrix.

Several notable relationships emerged from the analysis. Validator maturity variables exhibited the strongest correlations. The number of recorded epochs and the binary history indicator showed a strong positive correlation ($r = 0.87$), confirming that both variables capture aspects of validator longevity. Validator size, measured through log-transformed account balance, was also moderately correlated with maturity metrics ($r = 0.62$ with recorded epochs and $r = 0.57$ with history status), suggesting that larger validators tend to have longer operational histories.



Performance-related variables displayed weaker relationships than initially expected. Median efficiency percentage showed only moderate correlations with performance trend ($r = 0.37$) and validator history ($r = 0.39$), indicating that current performance and historical maturity represent distinct dimensions of validator behavior. Similarly, efficiency variability exhibited relatively weak correlations with the remaining features, suggesting that consistency captures information not explained by validator size, age, or performance level alone.

The absence of extremely strong correlations among most features indicated that each engineered variable contributed unique information. As a result, all seven features were retained for subsequent dimensionality reduction and clustering analysis.

4.2. Principal Component Analysis

Because the engineered features operate on different scales and may capture overlapping aspects of validator behavior, Principal Component Analysis (PCA) was applied prior to clustering. PCA reduces dimensionality while preserving the majority of variation present in the original feature space.

	PC	Explained Variance	Cumulative Variance
0	PC1	0.395647	0.395647
1	PC2	0.202149	0.597797
2	PC3	0.137844	0.735640
3	PC4	0.118623	0.854263
4	PC5	0.078588	0.932851
5	PC6	0.059138	0.991988
6	PC7	0.008012	1.000000

The first four principal components explained approximately 85.4% of the total variance in the dataset. Specifically, PC1 accounted for 39.6% of variance, PC2 explained 20.2%, PC3 explained 13.8%, and PC4 explained 11.9%. Because these

four components captured the majority of information while substantially reducing dimensionality, they were selected as the final representation used for clustering.

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
commission	-0.322725	0.007572	0.154699	0.885570	-0.110821	0.270575	0.046193
log_lambda	0.466965	-0.219110	-0.120437	-0.085929	-0.312395	0.783827	0.008315
median_efficiency_pct	0.317834	0.483863	-0.231219	0.178459	0.726909	0.220203	-0.051607
efficiency_cv	0.034246	0.521872	0.759998	-0.180527	-0.145366	0.167279	-0.259184
performance_trend	0.078295	0.628398	-0.482538	0.085041	-0.575547	-0.165373	0.017542
num_epochs_recorded	0.524076	-0.224866	0.044797	0.318914	-0.081684	-0.358786	-0.660050
has_history	0.542969	0.003218	0.309281	0.187100	-0.051704	-0.282588	0.701415

Interpretation of the component loadings revealed several meaningful latent dimensions of validator behavior.

PC1 (Validator Maturity and Scale) was driven primarily by validator history, recorded epochs, and account balance. Validators with high PC1 scores tended to be larger, more established operators with extensive performance histories.

PC2 (Performance Quality) was dominated by median efficiency and performance trend. High values along this dimension corresponded to validators exhibiting stronger operational performance and improving historical trajectories.

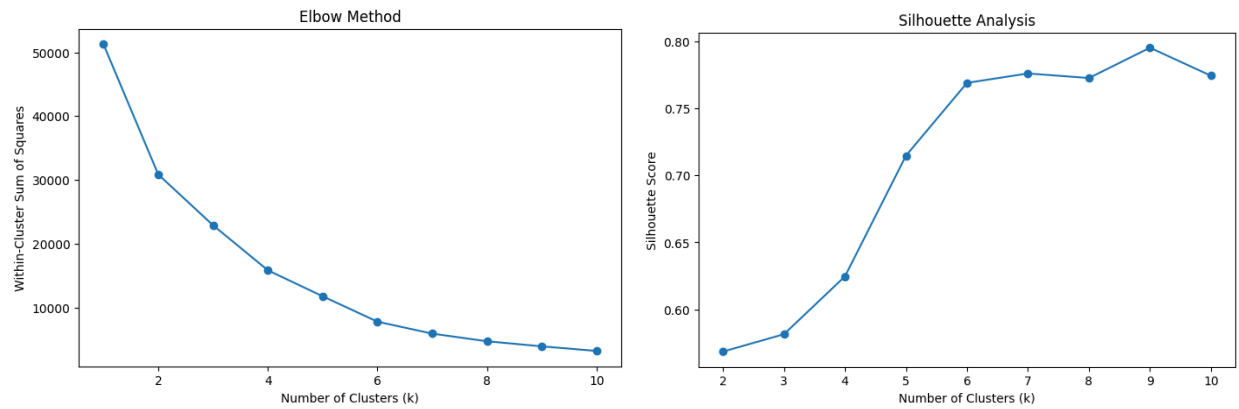
PC3 (Consistency and Stability) was largely influenced by efficiency variability. This dimension differentiated validators with stable performance from those exhibiting greater fluctuations across epochs.

PC4 (Commission Policy) was overwhelmingly driven by commission rate and therefore captured differences in validator fee strategies.

Together, these four components suggest that validator behavior can be understood primarily through four dimensions: scale, performance quality, consistency, and commission policy.

4.3. Cluster Selection

K-Means clustering requires the number of clusters, k , to be specified in advance. To determine an appropriate value, both the Elbow Method and Silhouette Analysis were evaluated across a range of cluster counts, as shown below.



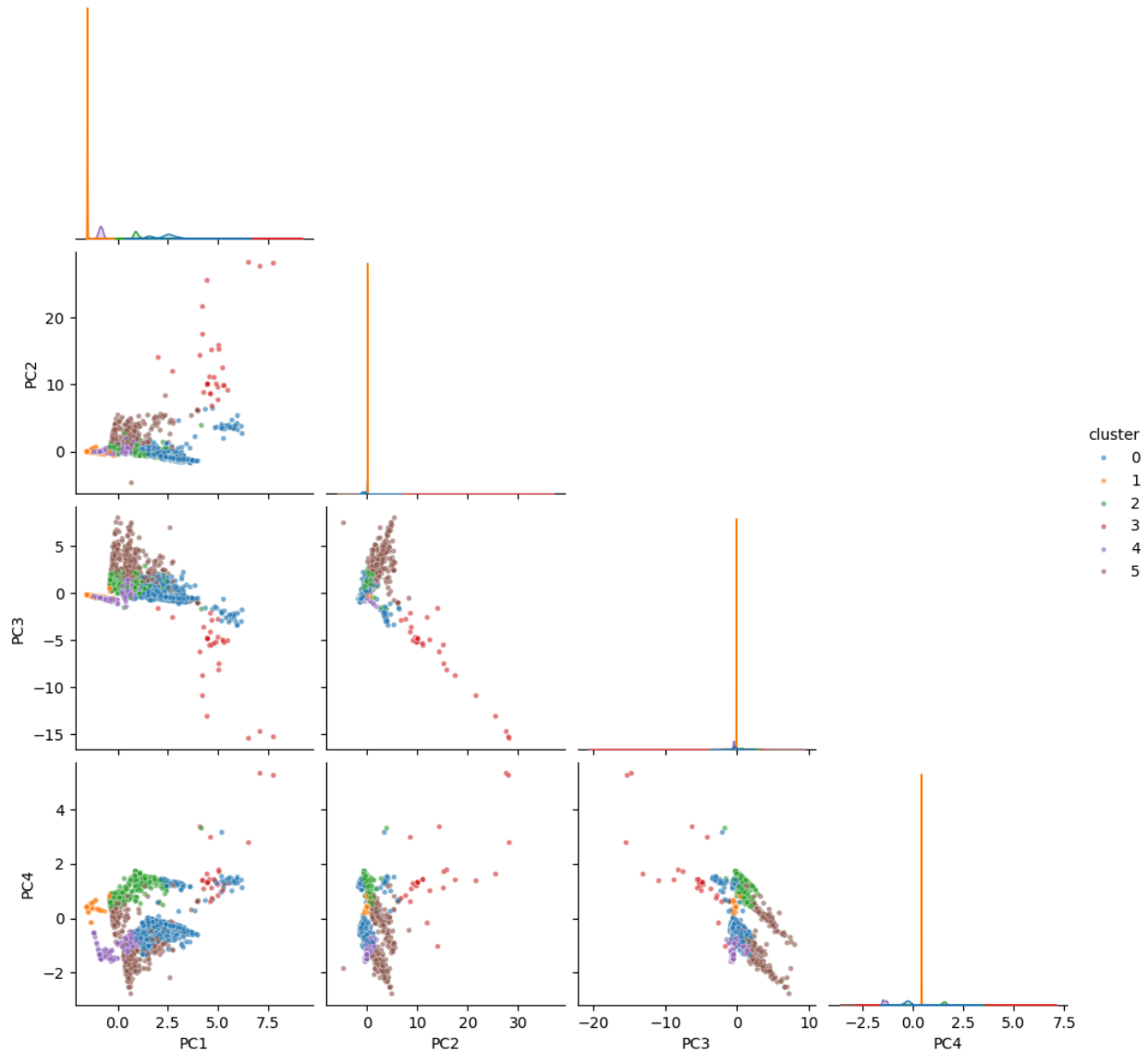
The Elbow Method showed a substantial reduction in within-cluster variance up to approximately six clusters, after which improvements diminished considerably. Silhouette scores increased steadily through $k=6$ and remained relatively stable beyond that point. Although slightly higher silhouette scores were observed for larger values of k , these gains were marginal and came at the cost of increased model complexity.

Based on the combined evidence from both metrics, six clusters were selected as a balance between cluster quality, interpretability, and model parsimony.

4.4. Validator Archetypes

After fitting the final K-Means model using six clusters, validators were projected into PCA space to visualize cluster separation, as shown on the following page.

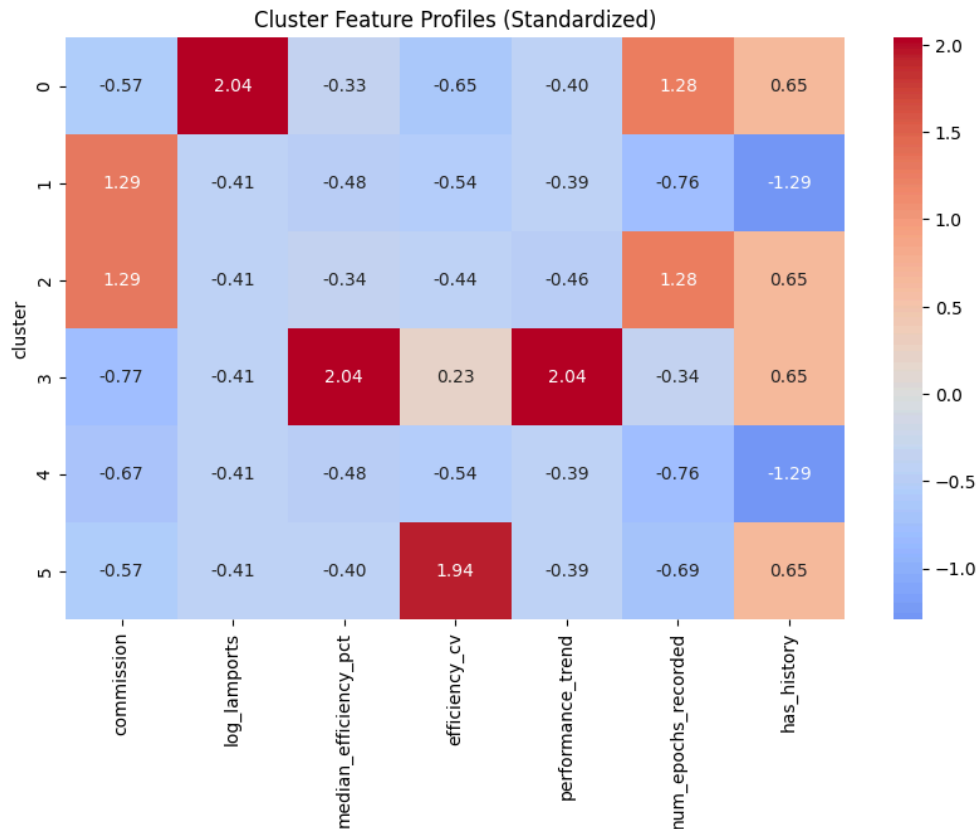
Pairwise PCA Space by Cluster



Several distinct patterns are visible in the PCA projections. While some overlap exists among clusters, clear separation emerges along multiple principal component combinations. Cluster 3 appears particularly isolated, occupying regions associated with exceptionally high performance scores. Clusters 1 and 4 are concentrated near the origin and represent validators with limited historical information. Other clusters separate primarily along dimensions related to maturity, consistency, and commission strategy.

The presence of distinguishable groups across several PCA dimensions suggests that Solana validators are not homogeneous participants but instead occupy identifiable regions within the operational landscape.

To better understand these groups, standardized cluster profiles were examined.



The cluster profiles reveal six distinct validator archetypes:

Cluster 0: Low-Commission Mature Validators (2,106 validators, 24.6%)

This cluster consists of large, long-established validators with extensive operational histories. Members maintain relatively low commission rates and exhibit strong performance consistency. These validators likely represent established operators prioritizing competitiveness and long-term stake attraction.

Cluster 1: Inactive Validators (3,379 validators, 39.4%)

Validators in this cluster exhibit very limited historical activity and almost no meaningful performance history. Despite their limited maturity, they maintain extremely high commission rates. This group may represent inactive, abandoned, or minimally utilized validator accounts.

Cluster 2: High-Commission Mature Validators (1,083 validators, 12.6%)

Like Cluster 0, these validators possess extensive operational histories. However, they differ by maintaining commission rates near the maximum observed values. The existence of this cluster suggests that some mature operators successfully employ significantly different economic strategies despite otherwise similar operational characteristics.

Cluster 3: High Performance Validators (35 validators, 0.4%)

Cluster 3 represents the most distinctive group in the dataset. Validators in this cluster exhibit exceptionally high efficiency scores and strongly positive performance trends. Although

relatively small in size, these validators appear to be the highest-performing operators within the network and occupy a clearly separated region in PCA space.

Cluster 4: Emerging Validators (1,625 validators, 18.9%)

This cluster contains newer validators with limited historical records and low commission rates. These operators may represent early-stage validators attempting to attract stake by minimizing fees while establishing a track record.

Cluster 5: Volatile Validators (348 validators, 4.1%)

Validators in Cluster 5 are characterized by unusually high efficiency variability relative to the rest of the network. Although many possess operational histories, their performance is substantially less consistent than other mature validators. This group may represent validators experiencing infrastructure instability, fluctuating stake levels, or other operational challenges.

Taken together, these results provide evidence that the Solana validator ecosystem contains several distinct operational archetypes rather than a single homogeneous population. Differences in maturity, performance, consistency, and commission strategy appear to drive much of the variation observed across validators, revealing meaningful structural patterns within the network.

5. Conclusion and Future Study

The objective of this project was to determine whether meaningful validator archetypes could be identified directly from on-chain Solana data using unsupervised learning techniques. By combining validator commission information, account balances, operational history, and performance-related metrics, a validator-level dataset was constructed and analyzed using Principal Component Analysis and K-Means clustering.

The results demonstrate that Solana validators do not behave as a homogeneous population. Instead, several distinct validator archetypes emerged from the data, including mature low-commission validators, mature high-commission validators, emerging validators, inactive validators, volatile validators, and a small group of exceptionally high-performing validators. These groups differed across multiple dimensions, including operational scale, longevity, commission strategy, performance quality, and performance consistency.

These findings are important because validators play a central role in maintaining the security, reliability, and decentralization of proof-of-stake blockchain networks. Understanding how validators differ provides insight into the structure of the Solana ecosystem and highlights the diversity of operational strategies employed by network participants. More broadly, this study demonstrates that statistical learning techniques can successfully uncover meaningful structure within large-scale blockchain datasets using only publicly available on-chain information.

This project also establishes a foundation for future research. While the current analysis focused on descriptive discovery through unsupervised learning, the identified validator archetypes naturally suggest several predictive research directions. Future work could investigate whether validator performance, reliability, or stake growth can be predicted using historical behavioral patterns. In particular, forecasting stake flows between validators represents a promising extension, as stake allocation directly influences validator rewards, network influence, and the long-term evolution of the ecosystem. The archetypes discovered in this study may also serve as labels or explanatory variables within supervised machine learning models designed to predict validator outcomes.

Beyond validator-level prediction, future research could explore forecasting overall network performance under varying conditions. Because validators collectively determine the throughput, reliability, and security of the Solana blockchain, predictive models may help estimate how network behavior changes in response to shifts in validator composition, stake distribution, participation rates, or other operational factors. Such models could provide valuable insights for network operators, researchers, and protocol developers seeking to improve network resilience and performance.

An additional direction involves stochastic modeling of validator and network dynamics. Validator behavior evolves continuously over time as operators gain or lose stake, modify commission policies, improve infrastructure, experience outages, or leave the network entirely. Rather than treating validators as static entities, stochastic processes could be used to model transitions between archetypes and quantify the uncertainty associated with future validator behavior. More broadly, stochastic modeling may provide a framework for understanding how network-level outcomes emerge from the interactions of thousands of individual validators operating under uncertain and constantly changing conditions.

Ultimately, this study represents a first step toward a broader research agenda focused on understanding blockchain ecosystems through data-driven methods. By identifying validator archetypes directly from on-chain activity, the project provides both practical insight into the current Solana network and a foundation for future predictive and stochastic modeling efforts aimed at understanding validator behavior, stake dynamics, and network performance.

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Appendix*

**Full notebook uploaded as separate file since it was too long when including all outputs.*